Reg No.:_____ Name:_____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: CHT205

Course Name: FLUID AND PARTICLE MECHANICS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Discuss the variation in viscosity of gases and liquids with temperature.	(3)
2	Give principle of inclined tube manometer.	(3)
3	Explain the significance of stream function.	(3)
4	Water flows through a channel, where the cross-sectional area of channel becomes 1.5 times the initial value. If the flow velocity in smaller cross section is 2 m/s, find out velocity in large cross section of the channel.	(3)
5	State Prandtl one seventh power law.	(3)
6	Water (density = 1g/cc and viscosity = 1 cP) flows through annulus formed between two concentric cylinders with diameters 5 cm (outer cylinder) and 4 cm (inner cylinder) respectively. If flow velocity is 1 cm/s, determine the type of flow.	(3)
7	Define form drag. How it can be minimized?	(3)
8	Comment on the statement "The generalization that liquids give particulate fluidization with solids while gases give bubbling fluidization is not completely valid".	(3)
9	Describe the concept of equivalent length.	(3)
10	How NPSH can be estimated?	(3)

PART B

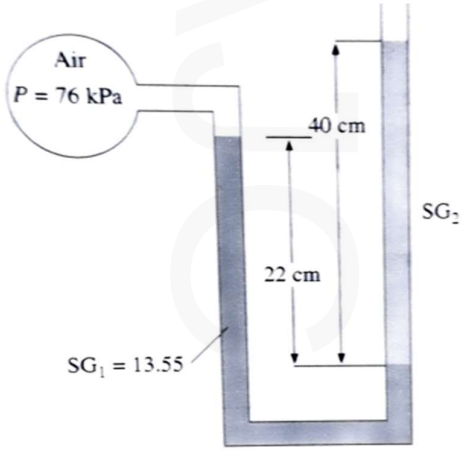
Answer any one full question from each module. Each question carries 14 marks

Module 1

11a	Calculate the capillary rise of a liquid (density = 1.26 g/cc) expected in a 0.8 mm diameter tube. Surface tension of the liquid with air = 64 mN/m, contact angle made by the liquid with tube = 60^0. What will be the rise if diameter of the tube is reduced to 0.5 mm?	(5)

b What is rheology? Describe rheological classification of fluids. (9)

12a A double fluid manometer is attached to an air pipe as shown. The specific gravity of one fluid is 13.55 and absolute pressure of air is 76 kPa. Find out the specific gravity for the other fluid, taking atmospheric pressure as 100 kPa. (5)



b Derive barometric equation. (9)

Module 2

13a What is boundary layer? Explain formation of boundary layer in flow over a flat plate. (7)

b Give Bernoulli's equation. How do you modify this equation in the presence of fluid friction and pump work? Explain. (7)

14a How do you classify flow based on Reynolds number? Explain the statistical nature of turbulence. (6)

b An oil is flowing through a pipe at a certain velocity. If this oil is replaced by another fluid having 0.85 times the density and 1.25 times the viscosity of the oil, what is the change in Reynolds number? Assume mass flow rate is kept constant. What is the change in Re if velocity is kept constant? (8)

Module 3

15a Derive the relationship between maximum and average velocity of flow through a circular pipe under laminar condition. Give all the assumptions used. (12)

b Define friction factor. How do you estimate friction factor for laminar flow in a channel? (2)

16a Explain the use of friction factor chart. (5)

b A fluid (specific gravity = 0.9 and viscosity = 3 cP) is draining from the bottom of a tank. The fluid level in the tank is 2 m. The drain pipe is of 0.2 m diameter (9)

and 10 m long. The flow rate in the drain pipe is 2 m^3/h and the line contains one elbow and two gate valves. Estimate the frictional loss in the drain pipe. Take loss coefficient for elbow and gate valve respectively as 0.9 and 0.2.

Module 4

17 Derive Ergun equation. (14)

18a What is superficial velocity? Discuss variation of pressure drop through a fluidization column varies with superficial velocity. (6)

b Air flows through a packed bed of a powdery material of 1 cm depth at a superficial gas velocity of 1 cm/s. A manometer connected to the unit registers a pressure drop of 1 cm of water. Assuming that Kozeny Carman equation is valid, estimate particle size of the powder. Take porosity of bed = 0.4, density of air = 1.23 kg/m^3 and viscosity of air = 1.8×10^{-5} kg/(m.s) (8)

Module 5

19a Describe the working principle of venturi meter. Derive flow rate equation for venturi meter. (10)

b Air is forced through a long circular channel of 0.9 m diameter. A pitot tube reading is taken at the centre, which is found to be 14 mm H_2O. Estimate the flow velocity in the channel, taking density of air as 1.29 kg/m^3. The coefficient of the pitot tube is 0.98. (4)

20a Explain characteristic curves for centrifugal pump. (10)

b The flow rate of water flowing in a 3 m wide channel is measured using a triangular weir, 0.5 m above the channel bottom, with a notch angle of 60^0. If the flow depth upstream of the weir is 1.5 m, determine the flow rate of water through the channel. Weir discharge coefficient = 0.6. (4)

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